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Year of Birth: 1994
Nationality: Mexican
Settlement Permit – Researcher



Carlos S. Reyna-Blanco, PhD

Personal Profile

Computational biologist with 10+ years of experience co-leading omics and biomedical data analysis projects. Expertise in developing reproducible bioinformatics pipelines and software tools, and in guiding teams to translate data into biological and clinical insights.

Experience

Aug 2024 – Present **Postdoc Researcher**, *Center for Cancer Research (Thomas Vogl lab)*, Vienna, Austria
- Developing scalable, reproducible statistical and machine learning pipelines for disease classification and survival analysis of high-throughput serological data, driving biomarker discovery across cohorts of >200 samples and >100k features.
- Partnering with clinicians and data scientists to support grant applications, streamline data workflows, and identify disease-associated immune signals in liver and bladder cancer studies, resulting in one first-author manuscript and multiple co-authored works under review.
- Mentoring PhD researchers in bioinformatics best practices, accelerating project delivery.

Sep 2023 – Dec 2023 **Data Scientist**, *Novartis Institutes for BioMedical Research*, Basel, Switzerland
- Deployed an R-based quality control pipeline for a clinical trial in prostate cancer, improving data reliability and standardization across samples.
- Led end-to-end processing and integration of single-cell RNAseq data across prostate cancer cohorts, facilitating characterization of tumor heterogeneity and cellular composition.

May 2018 – Jun 2023 **PhD Scholar in Bioinformatics**, *University of Fribourg (Daniel Wegmann lab)*, Switzerland
- Developed and extended C++ tools (TreeSwirl, ATLAS) by using machine learning and numerical methods to infer population genetic parameters, enabling analysis of low-quality genomic datasets.
- Optimized and deployed a Snakemake pipeline for processing of low-quality DNA samples, increasing data recovery and enabling reliable downstream population genomic analyses.

Dec 2016 – Mar 2018 **Bioinformatics Assistant**, *University of Bern (Anna Malaspinas lab)*, Bern, Switzerland
- Integrated public human genomic sequencing data to study population migrations.

Education

May 2018 – Jun 2023 **PhD in Bioinformatics**, *University of Fribourg*, Fribourg, Switzerland
Thesis: "Going with the (Gene) Flow: Computational tools for Inferring Population History from Genomic Signatures of Admixture." Graduated with highest honors. (Daniel Wegmann lab).

Sep 2016 – Feb 2018 **MSc in Bioinformatics and Computational Biology**, *Univ. of Bern*, Switzerland
Thesis: "Archaic Introgression in Australo-Papuan Populations." (Anna Malaspinas lab).

Skills

Programming	Python, R, C++, SQL, Bash
Tools/OS	Linux, Git, Docker, HPC/Slurm, Snakemake/Nextflow, scikit-learn, tidyverse, Jira
Languages	Spanish (native), English (fluent), French (intermediate), German (A2.2, basic proficiency)

Selected Software & Tools

phipml	Machine learning pipeline for disease classification using serological data (Python)
phipsurv	Survival modeling framework for prognostic analysis (Python)
phipreport	Automated reporting pipeline for large-scale serological profiling data (R, markdown)
TreeSwirl, ATLAS	Population genomics tools for low-quality and complex datasets (C++)

Selected Publications

2026 **Reyna-Blanco CS**, Scheiner B, ..., Vogl T *Deciphering the microbiota-immune axis in liver health and disease yields diagnostic and prognostic serological signatures* **Under review**

2025 Tsoupas A, **Reyna-Blanco CS**, Quilodrán CS, Blöcher J, Brami M, Wegmann D, Burger J, Currat M (2025) *Local increases in admixture with hunter-gatherers followed the initial expansion of Neolithic farmers across continental Europe*. ScienceAdvances

2024 **Reyna-Blanco CS**, ..., C, Wegmann D (2024) *Inference of Locus-Specific Population Mixtures from Linked Genome-Wide Allele Frequencies*. Molecular Biology and Evolution

Teaching

2019–2022 Machine Learning (Master's level, spring semester)
Teaching Assistant. Delivered lectures on Hidden Markov Models and led weekly exercises.

2019–2022 Introduction to Programming with R (Master's level, 3 weeks)
Teaching Assistant. Delivered lectures on data types, control flow, and tidyverse.

2019–2022 Introduction to UNIX and Bash Scripting (Master's level, 2 weeks)
Teaching Assistant. Delivered lectures and hands-on sessions on UNIX and Bash.

2018–2022 Bioinformatics and Genomics (Bachelor's level, fall semester)
Teaching Assistant. Led practical sessions and supervised student exercises.

Scholarships

2016 Awarded admission with funding to the Interdisciplinary Master's in Life Sciences (IMaLiS), École Normale Supérieure, Paris, France (declined).

2015 UNAM International Mobility, one year internship at the University of Leipzig, Germany.

2015 Globalink Research Internship, a summer research at Dalhousie University, Canada.

Soft Skills

Knowledge Translation	Communicate complex technical concepts to diverse audiences, including clinicians, biologists, and data scientists.
Problem Solving	Develop data-driven solutions to biological questions, troubleshoot pipelines, and adapt methods to challenging datasets.
Project Ownership	Manage multiple complex projects independently, balancing long-term research goals with timely deliverables.